Riemann Surfaces as One-Dimensional Complex Manifolds

N-20220212

§1.1 Structure beyond topology

The sphere and the torus are both smooth real surfaces, yet their meromorphic maps obey different global constraints. To see the source of the difference, begin with what local coordinates record. A topology determines which subsets are open; a smooth atlas makes differentiation possible; a complex atlas requires that every change of coordinates be holomorphic.[1]

Example 1.1.1

The interval $(0, 2) \subset \mathbb{R}$ is a topological 1-manifold. One chart may be the identity map

$$\varphi_1(x) = x.$$

The alternative coordinate $\psi(x) = 1 + \sqrt[3]{x - 1}$, using the real cube root, is also a homeomorphism from $(0, 2)$ onto itself. Yet its derivative is unbounded at $x = 1$. The transition map between these two charts fails to be C^1, so the two charts are not compatible in a smooth atlas. Topology alone cannot decide smooth differentiability.

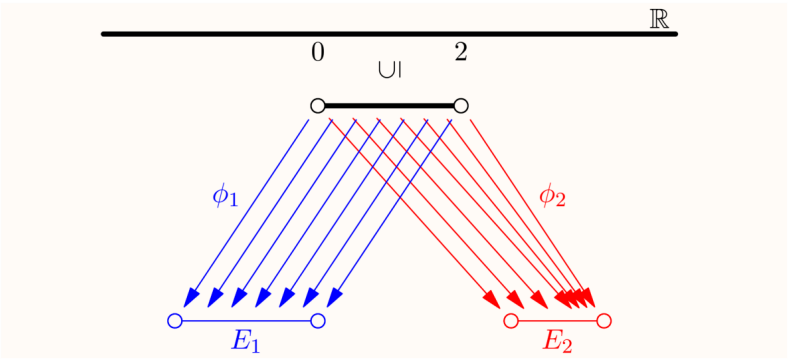


Figure 1.1: Two coordinate descriptions of the interval used to separate topology from smooth structure.

[1] The local normal form and branched-cover constructions follow O. Forster, *Lectures on Riemann Surfaces*. The topological proof of Riemann–Hurwitz is developed in E. Looijenga's Riemann-surface lectures.

Remark 1.1.2 — The lesson is that the same underlying topological space may admit different additional structures. Smoothness is not merely a property of the set; it is encoded by charts and transition maps.

§1.2 From smooth structure to complex structure

A smooth surface embedded in $\mathbb{R}^3$ can be described locally by diffeomorphisms to open subsets of $\mathbb{R}^2$. But this embedded definition is not sufficiently general: for example, the Klein bottle cannot be embedded in $\mathbb{R}^3$.

The abstract definition of a smooth manifold therefore uses charts

$$\varphi_i : U_i \to E_i \subseteq \mathbb{R}^n$$

whose transition maps are smooth. The same philosophy gives complex manifolds by replacing $\mathbb{R}^n$ with $\mathbb{C}^n$ and smooth transition maps with holomorphic transition maps.

Remark 1.2.1 — Because holomorphic functions are automatically analytic, a complex structure is very rigid. This is why a Riemann surface is much more than a real two-dimensional smooth manifold.

§1.3 Definition of a Riemann surface

Definition 1.3.1. A Riemann surface is a connected, Hausdorff, second-countable topological space X equipped with an atlas $\{(U_i, \varphi_i)\}$, where

$$\varphi_i : U_i \xrightarrow{\sim} E_i \subseteq \mathbb{C}$$

is a homeomorphism onto an open subset of $\mathbb{C}$, and all transition maps

$$\varphi_j \circ \varphi_i^{-1}$$

are holomorphic on their domains.

Each φ_i is a complex chart. A local coordinate near $p \in X$ is the complex number

$$z = \varphi_i(x).$$

If $\varphi_i(p) = 0$, the coordinate is centered at p.

Definition 1.3.2. More generally, a complex n-manifold is a Hausdorff space locally modeled on open subsets of $\mathbb{C}^n$, with holomorphic transition maps.

Every complex n-manifold is naturally a smooth real $2n$-manifold.

§1.4 Examples

Example 1.4.1 (Open subsets of the plane)
Every connected open subset $U \subseteq \mathbb{C}$ is a Riemann surface, with the identity chart.

Example 1.4.2 (The Riemann sphere)

The Riemann sphere $\mathbb{C}_\infty$ is the complex plane together with a point at infinity. One way to construct its complex structure is to cover it by two charts: one chart with coordinate z, and another chart near infinity with coordinate

$$t = 1/z.$$

On the overlap, the transition function is

$$z = 1/t,$$

which is holomorphic away from 0. Thus the two charts glue into a Riemann surface.

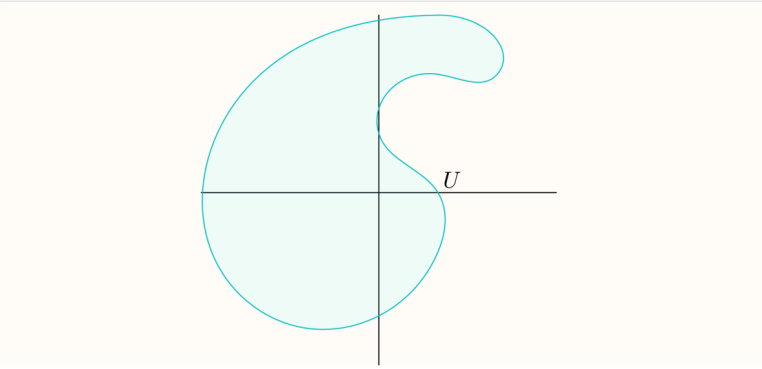


Figure 1.2: Stereographic charts used to define the complex structure on the Riemann sphere.

Caution

The sign choice in the stereographic chart matters. If the transition map became $z = 1/\bar{t}$, it would be antiholomorphic rather than holomorphic. The complex structure depends on getting this orientation right.

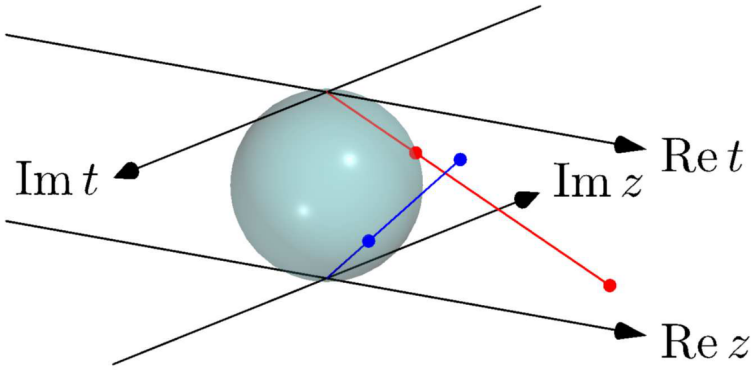


Figure 1.3: The second stereographic chart and the transition coordinate near infinity.

Example 1.4.3 (A complex torus)
Let $L = \mathbb{Z}[i] \subset \mathbb{C}$, a lattice under addition. The quotient

$$\mathbb{C}/L$$

is a complex torus. Locally, the quotient map from $\mathbb{C}$ gives complex coordinates, and different choices of lifts differ by translations

$$z \mapsto z + a, \qquad a \in L,$$

which are holomorphic.

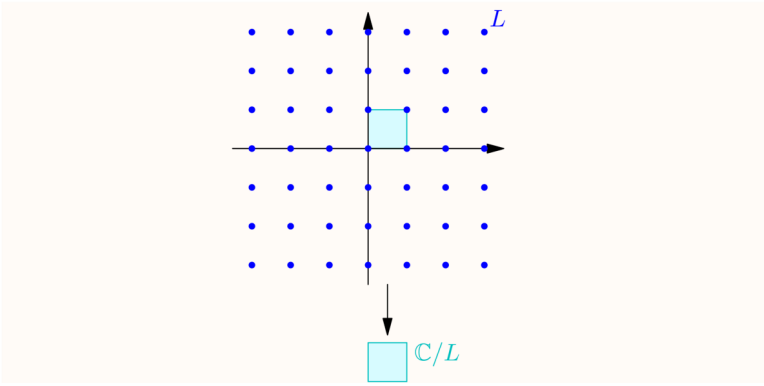


Figure 1.4: The quotient picture of a complex torus, with opposite sides identified.

Remark 1.4.4 — The complex torus is compact. Any holomorphic function on it is constant, but meromorphic functions on it are much more interesting.

Example 1.4.5
A disjoint union of two Riemann spheres is not a Riemann surface under the convention used here, because a Riemann surface is required to be connected. If one drops connectedness, each connected component is a Riemann surface.

§1.5 Holomorphic maps between Riemann surfaces

Definition 1.5.1. Let X, Y be Riemann surfaces. A map $f : X \to Y$ is holomorphic at $p \in X$ if, for charts

$$\varphi_1 : U_1 \to E_1, \qquad \varphi_2 : U_2 \to E_2$$

with $p \in U_1$ and $f(p) \in U_2$, the local expression

$$\varphi_2 \circ f \circ \varphi_1^{-1}$$

is holomorphic near $\varphi_1(p)$. The map is holomorphic if this holds at every point.

Example 1.5.2 (The cubic map)

The map

$$f : \mathbb{C} \to \mathbb{C}, \qquad f(z) = z^3$$

is holomorphic. In the usual global coordinate on $\mathbb{C}$, this is just a polynomial, hence holomorphic everywhere. At a point $a \neq 0$, the derivative

$$f'(a) = 3a^2$$

is nonzero, so the inverse function theorem gives a local holomorphic inverse near a. At 0, however, the derivative vanishes. The local model is exactly

$$z \mapsto z^3,$$

so the map has multiplicity 3 at the origin. This is the first concrete place where "holomorphic map" and "branched covering" begin to touch.

Example 1.5.3 (Inclusion into the Riemann sphere)

The inclusion

$$\mathbb{C} \hookrightarrow \mathbb{C}_\infty$$

is a holomorphic map of Riemann surfaces. On the finite chart of $\mathbb{C}_\infty$, this map is simply

$$z \mapsto z,$$

so it is holomorphic in the most literal possible sense. The point of the example is not the formula itself, but the target: the Riemann sphere adds one extra point ∞, and the ordinary plane sits inside it as the finite chart. Thus a map into $\mathbb{C}_\infty$ may be checked by using the coordinate z away from infinity and the coordinate $w = 1/z$ near infinity.

§1.6 Meromorphic functions as maps to the sphere

The Riemann sphere lets us treat poles as honest values. A meromorphic function

$$f : X \to \mathbb{C}$$

can be regarded as a holomorphic map

$$g : X \to \mathbb{C}_\infty$$

by sending poles to ∞.

Proposition 1.6.1

Nonconstant meromorphic functions $X \to \mathbb{C}$ correspond to holomorphic maps $X \to \mathbb{C}_\infty$ not identically equal to ∞.

Proof. Near a pole p, take a local coordinate z with $z(p) = 0$. A meromorphic function has the form $f(z) = z^{-m}h(z)$, where $m \geq 1$ and h is holomorphic with $h(0) \neq 0$. In the

target coordinate $t = 1/w$ near infinity, the composite is

$$t \circ f(z) = \frac{z^m}{h(z)},$$

which is holomorphic and sends p to $t = 0$. Thus f extends as a holomorphic map into the sphere. Conversely, for a nonconstant holomorphic map to the sphere, use $t = 1/w$ near a preimage of infinity. The function $t \circ f$ is holomorphic with an isolated zero of finite order m, so f has a pole of order m. At all other points it is an ordinary holomorphic function. □

§1.7 Degree and multiplicity

For nonconstant holomorphic maps between compact Riemann surfaces, fibres behave surprisingly well when counted with multiplicity.

Proposition 1.7.1

Let X, Y be compact Riemann surfaces and let $f : X \to Y$ be nonconstant holomorphic. For each $y \in Y$, count the points in $f^{-1}(y)$ with multiplicity. The resulting number is finite and independent of y. This number is the degree of f, denoted $\deg(f)$.

Remark 1.7.2 — This is much stronger than what one expects from arbitrary smooth maps. The rigidity of holomorphic maps makes the fibre count locally constant, and connectedness makes it globally constant.

Example 1.7.3 (The power map has degree k)

The map

$$z \mapsto z^k$$

extends to a holomorphic map $\mathbb{C}_\infty \to \mathbb{C}_\infty$ of degree k. On the finite chart this is a polynomial. Near infinity, use the coordinate $w = 1/z$. Then

$$z^k = \frac{1}{w^k}, \qquad \frac{1}{z^k} = w^k,$$

so the extended map is also holomorphic at ∞. For a generic value $a \in \mathbb{C}^*$, the equation

$$z^k = a$$

has k distinct solutions. Counting multiplicity, the same number remains true over the exceptional values 0 and ∞. This is why the degree is k, not merely because the formula contains the exponent k.

Definition 1.7.4. Let $f : X \to Y$ be a nonconstant holomorphic map and $p \in X$. The multiplicity of f at p is the unique integer $m \geq 1$ such that, after choosing suitable local coordinates centered at p and $f(p)$, the map is locally of the form

$$z \mapsto z^m.$$

Remark 1.7.5 — The guiding slogan is: every nonconstant holomorphic map locally looks like a power map. This is one of the reasons Riemann surfaces are so much cleaner than arbitrary smooth surfaces.

Idea of the local normal form. Choose local coordinates centered at p and $f(p)$, so the local expression becomes a nonconstant holomorphic function F with $F(0) = 0$. Its Taylor expansion has the form

$$F(z) = a_m z^m + a_{m+1} z^{m+1} + \cdots, \qquad a_m \neq 0.$$

Thus

$$F(z) = z^m u(z),$$

where u is holomorphic and $u(0) \neq 0$. After shrinking the coordinate neighborhood, u has a holomorphic m-th root v. Replacing the source coordinate z by $zv(z)$, the map becomes exactly

$$z \mapsto z^m.$$

The integer m is uniquely determined by the order of vanishing of F at 0, hence does not depend on the chosen coordinates. □

§1.8 Local power form is not global

The local normal form $z \mapsto z^m$ must be read with the word *local* emphasized. A useful test case is

$$U = \mathbb{C}^*, \qquad F : U \to \mathbb{C}^*, \qquad F(z) = z^2.$$

At $p = 1$, the derivative is nonzero, so in the local-normal-form sense the multiplicity is $m = 1$. Nearby, one can choose a local inverse

$$z = \sqrt{w}$$

with $w = F(z)$. But this square-root coordinate cannot be chosen globally on $\mathbb{C}^*$. If the target point travels once around the loop

$$w(t) = e^{2\pi i t}, \qquad 0 \leq t \leq 1,$$

then analytic continuation of the local branch starting at $\sqrt{1} = 1$ ends at -1, not 1. This sign change is monodromy.

There is a small trap here. If one instead moves the source point as

$$p(t) = e^{2\pi i t},$$

then $F(p(t)) = e^{4\pi i t}$ winds twice around the target origin, so the square root changes sign twice and returns to the original branch. The monodromy is seen by following a local inverse over a loop in the target, not by confusing the covering coordinate with a single global coordinate downstairs.

Thus the map $z \mapsto z^2$ is an unbranched two-fold covering of $\mathbb{C}^*$, while the missing point 0 is where the corresponding map on $\mathbb{C}$ would branch. To make $\sqrt{w}$ single-valued one must either cut the base, for example by removing a ray, or pass to a covering space. This is the global meaning of the failure: local coordinate charts glue with a monodromy representation, and on a non-simply-connected region those gluings need not collapse to one global coordinate.

§1.9 Why a compact surface forces meromorphic functions

The earlier examples distinguish the two basic compact Riemann surfaces: the sphere $\mathbb{P}^1(\mathbb{C})$, assembled from the z- and $1/z$-charts, and a complex torus $\mathbb{C}/\Lambda$, assembled from translations by a lattice. On either compact surface a global holomorphic function $f : X \to \mathbb{C}$ must be constant. Indeed, $|f|$ attains its maximum, and the maximum-modulus principle forces f to be constant on a connected surface.

Meromorphic functions avoid this obstruction by allowing isolated poles. A nonconstant meromorphic f determines a holomorphic map

$$f : X \longrightarrow \mathbb{P}^1(\mathbb{C}),$$

where each pole maps to ∞. Thus the poles are not merely defects of a function: they are precisely the points where the image enters the second coordinate chart on the sphere. For compact X, the resulting map has a finite degree d, and the local power map $z \mapsto z^{e_p}$ records its ramification index at p.

We can now ask a concrete topological question. If an analytic map has degree d and branches at some points, how can the number of handles of its source and target differ? The answer follows from counting cells over a covering.

§1.10 Riemann–Hurwitz by lifting a triangulation

Let $f : X \to Y$ be a nonconstant holomorphic map between connected compact Riemann surfaces, of degree d. Put

$$R(f) = \sum_{p \in X} (e_p - 1).$$

Only finitely many terms are nonzero: a nonconstant holomorphic map is locally a power map, and its critical points are isolated; compactness makes the total critical set finite.

Choose a triangulation of Y with every branch value among its vertices. Over an open face or an open edge, the map is an ordinary d-sheeted covering. Each open face and edge therefore has exactly d lifts. Over a vertex y, the number of distinct preimages is

$$\#f^{-1}(y) = d - \sum_{p \in f^{-1}(y)} (e_p - 1),$$

because the multiplicities of the points in the fiber sum to d.

Count Euler characteristics in the lifted triangulation:

$$\begin{aligned} \chi(X) &= V_X - E_X + F_X \\ &= dV_Y - R(f) - dE_Y + dF_Y \\ &= d\chi(Y) - R(f). \end{aligned}$$

Since an orientable compact surface of genus g has $\chi = 2 - 2g$, this yields

$$\boxed{2g_X - 2 = d(2g_Y - 2) + R(f).}$$

This is the Riemann–Hurwitz formula. The correction $R(f)$ counts how many distinct preimages of vertices are lost when several sheets merge at a ramification point. The proof explains why the local analytic multiplicities have a global topological cost.

§1.11 The power map checks every term

Take

$$f : \mathbb{P}^1 \to \mathbb{P}^1, \qquad z \longmapsto z^n \quad (n \geq 2).$$

A generic nonzero finite value has n distinct preimages, so $d = n$. At $z = 0$, the map has ramification index n. At infinity, use $w = 1/z$ in the source and $v = 1/f(z)$ in the target. Then $v = w^n$, so the index at infinity is also n. All other points have $e_p = 1$.

Both surfaces have genus zero, and the formula reads

$$-2 = n(-2) + (n-1) + (n-1) = -2.$$

The ramification at infinity is essential. Forgetting it would produce an apparent contradiction, precisely because an affine calculation sees only one of the two exceptional values.

§1.12 A square root of a polynomial produces a torus

Consider the compactification of the smooth affine curve

$$X : \quad y^2 = (x - a_1)(x - a_2)(x - a_3)(x - a_4),$$

where the complex numbers $a_1, \ldots, a_4$ are distinct. The projection

$$\pi : X \longrightarrow \mathbb{P}^1, \qquad (x, y) \longmapsto x$$

has degree two: generically there are two possible square roots.

At $x = a_i$, write $x = a_i + t^2$ in a local parameter t. Since the other three factors are nonzero there, the equation determines y as t times a nonvanishing holomorphic function. The projection is locally $t \mapsto t^2$, so each of the four points above a_i has $e = 2$.

We must still inspect infinity. Put

$$u = \frac{1}{x}, \qquad v = \frac{y}{x^2}.$$

Then the curve equation becomes

$$v^2 = \prod_{i=1}^{4} (1 - a_i u).$$

At $u = 0$, the equation has two distinct solutions $v = \pm 1$. They are regular points, and u is a local coordinate near either of them; hence the double cover is unramified over infinity.

The total ramification is therefore $R(\pi) = 4$, and Riemann–Hurwitz gives

$$2g_X - 2 = 2(-2) + 4 = 0, \qquad \boxed{g_X = 1.}$$

The smooth compactification is topologically a torus. More generally, if the defining polynomial has $2m$ distinct roots, the same computation gives $R = 2m$ and

$$2g_X - 2 = -4 + 2m, \qquad g_X = m - 1.$$

An algebraic square-root equation has acquired a precise topological genus.

§1.13 Why a Torus Differs from a Sphere

The two principal examples now behave differently even before one studies their geometry explicitly. A degree-one nonconstant holomorphic map between compact Riemann surfaces is a biholomorphism: the sum of multiplicities in each fiber is one, so every fiber has one point with local index one, and the map is a bijective local holomorphic isomorphism. In particular, a genus-one surface cannot map biholomorphically to the sphere.

The genus-one double cover above must have degree at least two; its four simple branch points make that possible. Conversely, a degree-two meromorphic function on a torus records the torus as two copies of the sphere joined along branch cuts. The analytic data of sheets and critical points and the topological data of handles are linked by a single cell-counting identity.

This is the utility of treating complex structure as something encoded by coordinate changes. Locally a nonconstant holomorphic map is only a power map. Globally those local powers may connect sheets in a way that produces a new surface, and the amount of branching measures the resulting change in Euler characteristic.